

COMP3069 – Computer Graphics

Practice Quiz (Lectures 1–5)

Duration: 30 minutes Marks: 100

Part A – Multiple Choice (2 marks × 10 = 20 marks)

1. Which of the following best describes rendering?

Answer: b) Converting geometric data into coloured pixels on screen

2. In OpenGL, the process of determining which pixels lie inside a triangle is called _____.

Answer: c) Rasterisation

3. The memory region that stores pixel colour values is known as _____.

Answer: b) Frame buffer (colour buffer)

4. The vertex shader _____.

Answer: b) Processes each vertex (e.g. position transformation)

5. Which library is used to create a window in an OpenGL program?

Answer: b) GLFW

6. A VBO (Vertex Buffer Object) _____.

Answer: b) Contains vertex attribute arrays in GPU memory

7. The dot product of two perpendicular (orthogonal) unit vectors equals _____.

Answer: c) 0

8. The cross product of two vectors produces _____.

Answer: c) A new vector perpendicular to both inputs

9. In a 4×4 transformation matrix, which component enables translation?

Answer: c) The fourth column (x,y,z offsets)

10. In combined transformations, the order of matrix multiplication is read _____.

Answer: b) Right → Left

Part B – Short Answer (5 marks × 4 = 20 marks)

1. Briefly describe the four stages of the 2D rendering pipeline.

Answer: 1. Vertex Specification – define vertices.

2. Vertex Shader – transform vertices.

3. Rasterisation – convert triangles to pixels.

4. Fragment Shader – compute pixel colours.

2. Explain the difference between a vertex shader and a fragment shader.

Answer: Vertex shader processes each vertex's position; fragment shader computes each pixel's colour.

3. What is a homogeneous coordinate, and why do we use it in transformations?

Answer: A 4th coordinate (w) added to 3D vectors, allowing translation to be represented in matrix form.

4. What are barycentric coordinates, and how can they be used to test if a point is inside a triangle?

Answer: They express a point as a weighted combination of triangle vertices; if all weights $\in [0,1]$, the point is inside.

Part C – Calculations (15 marks $\times$ 2 = 30 marks)

1. Given vectors $u = (2, 1, 2)$ and $v = (1, 0, -1)$:

a) $u + v = (3, 1, 1)$

b) $2u = (4, 2, 4)$

c) $\|u\| = \sqrt{2^2+1^2+2^2}=3$

d) $u \cdot v = 2 \cdot 1 + 1 \cdot 0 + 2 \cdot (-1) = 0$

e) Since $u \cdot v = 0$, the angle $\theta = 90^\circ$ (vectors are orthogonal).

2. Triangle $A(0,0)$, $B(2,0)$, $C(1,2)$, $P(1,1)$:

Using barycentric formula:

$\alpha=0.25$, $\beta=0.25$, $\gamma=0.5$

Since all $\alpha, \beta, \gamma \in [0,1]$ and sum to 1 $\rightarrow P$ is inside the triangle.

Part D – Application & Conceptual (15 marks × 2 = 30 marks)

1. Transformation matrices:

Translation(+2,+1): $\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Rotation(90°): $\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Scaling(2,0.5): $\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Combined (Rotation → Scaling → Translation): $M = T \times S \times R$.

2. Rendering workflow:

- VBO stores vertex data in GPU memory.
 - VAO describes how vertex data is structured.
 - Vertex shader transforms vertex positions.
 - Fragment shader colours fragments.
- Together they render a coloured triangle.